Comprehensive Multi-Sport Quantitative Analytics Platform: Serverless DuckLake Architecture, Latent State-Space Projections, and Cross-Sport Econometric Modeling

Master Architectural Blueprint & Empirical Research Specification (MLB → NBA → NHL → NFL)

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1 Phase 1: Major League Baseball — Kinematics, State-Space Projections, Pitch Modeling, and SABR Research

1.1 Statcast 9-Parameter Kinematics and the 2026 ABS Coordinate Shift

High-precision pitch evaluation requires reconstructing the three-dimensional flight trajectory of a baseball from optical and radar tracking telemetry operating at 20,000 frames per second (Ref. [A1]). In the Statcast coordinate system, spatial measurements are oriented from the catcher's perspective: the x-axis runs horizontally (positive to the catcher's right), the y-axis extends from the back apex of home plate toward the pitcher's mound, and the z-axis measures vertical height above the ground in feet (Ref. [A1, A2, A3]). While pitch velocities from 2008 through 2016 were captured via PITCHf/x and mathematically adjusted to approximate the out-of-hand release point, all Statcast velocities from 2017 onward (`release_speed`) are measured directly out-of-hand, placing historical and modern velocity distributions on a unified scale (Ref. [A2]).

To reconstruct the continuous trajectory of any pitch, the analytics engine evaluates the nine kinematic parameters initialized at the reference plane $y_0 = 50$ ft from the apex of home plate: initial position (x_0, y_0, z_0), initial three-dimensional velocity (v_{x0}, v_{y0}, v_{z0}) in feet per second, and constant three-dimensional acceleration (a_x, a_y, a_z) in ft/s^2 (Ref. [A1, A2, A3]). Under constant acceleration kinematics, the spatial position of the baseball at elapsed flight time t (relative to $y_0 = 50$ ft) is governed by the system of equations (Ref. [A1]):

$$x(t) = x_0 + v_{x0}t + \frac{1}{2}a_x t^2 \tag{1}$$

$$y(t) = y_0 + v_{y0}t + \frac{1}{2}a_y t^2 \tag{2}$$

$$z(t) = z_0 + v_{z0}t + \frac{1}{2}a_z t^2 \tag{3}$$

A critical structural break occurs in the 2026 Major League Baseball season that invalidates unadjusted multi-year spatial models. Through the 2025 season, Statcast recorded `plate_x` and `plate_z` at the front plane of home plate ($y_{\text{front}} = \frac{17}{12}$ ft ≈ 1.4167 ft forward of the back apex), while the vertical strike-zone boundaries (`sz_top` and `sz_bot`) were manually set by a human video tracking operator when the ball was halfway to the plate, historically adding four inches above the batter's belt for `sz_top` and marking the hollow of the knee for `sz_bot` (Ref. [A2, A3]). Beginning in the 2026 season, Statcast redefined `plate_x` and `plate_z` to capture the ball at the middle of home plate ($y_{\text{ABS}} = \frac{8.5}{12}$ ft ≈ 0.7083 ft) to align with the Automated Ball-Strike (ABS) challenge and tracking system, and transitioned `sz_top` and `sz_bot` to the batter's standardized ABS-defined biomechanical strike zone (Ref. [A2]).

Because a pitched baseball drops steeply over the final 8.5 inches ($\Delta y = -0.7083$ ft) between the front of the plate and the middle of the plate, combining raw 2020–2025 data with 2026+ data introduces systematic vertical and horizontal coordinate drift. To normalize all historical pitches (pre-2026) to the 2026 ABS middle-of-plate plane ($y_{\text{target}} = 0.7083$ ft), the ingestion pipeline solves the quadratic trajectory equation for the exact flight time t_{ABS} required to travel from $y_0 = 50$ ft to y_{target} (Ref. [A1, A2]):

$$t_{\text{ABS}} = \frac{-v_{y0} - \sqrt{v_{y0}^2 - 2a_y(y_0 - y_{\text{target}})}}{a_y} \tag{4}$$

Substituting t_{ABS} into the $x(t)$ and $z(t)$ equations yields the ABS-normalized coordinates ($\text{plate_x}_{\text{ABS}}$, $\text{plate_z}_{\text{ABS}}$) for all pre-2026 seasons (Ref. [A1, A2]). Furthermore, spin-induced movement parameters (`pfx_x` and `pfx_z`), which measure the horizontal and vertical deflection in feet at $y = 40$ ft relative to a theoretical spinless pitch thrown at the same velocity (including drag), are standardized into pitcher-perspective inches via the transformation (Ref. [A2, A3, A4, A5, A6]):

$$\text{hMov}_{\text{in}} = -12 \cdot \text{pfx_x}, \quad \text{vMov}_{\text{in}} = 12 \cdot \text{pfx_z} \tag{5}$$

Under this convention, pitches with pure backspin (spin_axis = 180°) exhibit positive induced vertical break ($\text{vMov}_{\text{in}} > 0$), whereas pure topspin 12-6 curveballs (spin_axis = 0°) exhibit negative induced vertical break (Ref. [A1, A2, A4]).

Parameter	Unit & Frame	Pre-2026 Legacy	2026+ ABS Standard	Pipeline Normalization
`plate_x`	Feet (Catcher $+x$ right)	Front of plate ($y = 1.4167$ ft) (Ref. [A2])	Middle of plate ($y = 0.7083$ ft) (Ref. [A2])	Project pre-2026 rows to $y = 0.7083$ ft via $x(t_{\text{ABS}})$ (Ref. [A1, A2])
`plate_z`	Feet above ground ($+z$ up)	Front of plate ($y = 1.4167$ ft) (Ref. [A2])	Middle of plate ($y = 0.7083$ ft) (Ref. [A2])	Project pre-2026 rows to $y = 0.7083$ ft via $z(t_{\text{ABS}})$ (Ref. [A1, A2])
`sz_top` / `sz_bot`	Feet above ground	Manual video operator line at $y \approx 25$ ft (Ref. [A2, A3])	Automated ABS biomechanical zone (Ref. [A2])	Normalize height: $z_{\text{norm}} = \frac{\text{plate_z}_{\text{ABS}} - \text{sz_bot}}{\text{sz_top} - \text{sz_bot}}$ (Ref. [A2])
`pfx_x` / `pfx_z`	Feet at $y = 40$ ft	Spin-induced break in feet (Ref. [A2, A4, A5])	Spin-induced break in feet (Ref. [A2])	Multiply by -12 (`pfx_x`) and $+12$ (`pfx_z`) for pitcher inches (Ref. [A4, A6])
`arm_angle`	Degrees (0° horiz.)	Shoulder-to-ball release vector angle (Ref. [A2])	Shoulder-to-ball release vector angle (Ref. [A2])	Retain scale; mirror `release_pos_x` for LHP parity (Ref. [A2, A7])
`swing_path_tilt`	Degrees vs. ground	40 ms pre-contact flat-disc swing plane (Ref. [A2])	40 ms pre-contact flat-disc swing plane (Ref. [A2])	Pair with `attack_angle` and `attack_direction` for CGD (Ref. [A2, A8])

1.2 Stuff+, Location+, and Pitching+ Architecture

To isolate intrinsic pitch quality (Stuff+) from spatial execution (Location+) and overall pitch value (Pitching+), the platform trains stratified gradient-boosted decision tree ensembles (`LightGBM`, `CatBoost`, and `HistGradientBoosting`) across 700,000+ pitch-level Statcast observations per season (Ref. [A7, A9, A10, B61, B62]). Rather than training directly on binary strikeout or noisy ERA outcomes, the target variable is constructed from the ground up using pitch-level linear weight run values (ΔRE24), where balls, called strikes, swinging strikes, fouls, and balls in play (valued via expected weighted on-base average, `xwOBA`) are assigned context-neutral run values (Ref. [A7, A10, A11]).

Pitches are partitioned into three primary classification families—Fastballs (Four-Seam, Sinker, Cutter), Breaking Balls (Slider, Sweeper, Curveball, Knuckle-Curve), and Offspeed (Changeup, Splitter)—yielding 12 distinct sub-models when crossed with four algorithmic specifications (Ref. [A7]). To prevent data leakage where the tree ensemble memorizes individual pitcher identities via release coordinates, hyperparameter tuning via Optuna enforces pitcher-grouped K-fold cross-validation, ensuring all pitches thrown by pitcher i reside exclusively in either the training fold or the validation fold (Ref. [A10]). Left-handed pitcher horizontal features (`release_pos_x`, `pfx_x`, `vx0`, `ax`) and `spin_axis` ($\theta_{\text{norm}} = 360° - \text{spin_axis}$) are reflected across the vertical axis prior to training so physical movement profiles remain handedness-invariant (Ref. [A2, A7]).

For Secondary and Tertiary pitches (Breaking Balls and Offspeed), physical deception depends heavily on tunneling relative to the pitcher's primary fastball. Accordingly, the Stuff+ feature matrix incorporates both absolute kinematics (`release_speed`, `release_extension`, `release_pos_x`, `release_pos_z`, `arm_angle`, `release_spin_rate`, `spin_axis`, `pfx_x`, `pfx_z`) and differential features ($\Delta v = v_{\text{pitch}} - \bar{v}_{\text{FB}}$, $\Delta\text{pfx_x} = \text{pfx_x}_{\text{pitch}} - \overline{\text{pfx_x}}_{\text{FB}}$, $\Delta\text{pfx_z} = \text{pfx_z}_{\text{pitch}} - \overline{\text{pfx_z}}_{\text{FB}}$) (Ref. [A2, A6, A10]). Conversely, Location+ strips out all velocity, movement, and spin variables, training exclusively on count state (balls, strikes), platoon handedness, and ABS-normalized spatial coordinates ($\text{plate_x}_{\text{ABS}}, z_{\text{norm}}$) (Ref. [A2, A7, A12]). Raw model predictions $\hat{y}$ are standardized onto the conventional plus-scale where league average equals 100 and one standard deviation equals 10 (Ref. [A11]):

$$\text{Stuff+}_p = 100 - 10 \cdot \left(\frac{\hat{y}_p - \mu_{\hat{y}}}{\sigma_{\hat{y}}} \right) \tag{6}$$

Empirical validation demonstrates that while FanGraphs' proprietary Stuff+ model retains a narrow advantage in forecasting full-season ERA ($r = -0.358$ vs. -0.317, a differential of 0.041), an Optuna-tuned LightGBM Stuff+ pipeline trained on pitch-level xwOBA and run-value frameworks significantly outperforms FanGraphs Stuff+ on direct bat-missing indicators, achieving a +0.514 correlation with Whiff Rate (compared to +0.355 for FanGraphs, a +0.159 correlation advantage) and a +0.417 correlation with Strikeout Rate (compared to +0.390 for FanGraphs, a +0.027 advantage) (Ref. [A10]). Furthermore, across 1.76 million pitches in in-house validation, pitch-level physical metrics correlate with next-season major league strikeout rate within 100 to 200 pitches, even

while pitch-level raw run values remain noisy ($r = .021$), and fastball velocity and Stuff+ stabilize within three to five appearances (Ref. [B4, B61, B63]).

1.3 Quantifying Pitcher Intent and Execution via xCTRL Gaussian Mixture Models

A fundamental limitation of Location+ and traditional walk-based control metrics is that they evaluate where a pitch crossed the plate relative to a league-wide run-value map, entirely ignoring where the pitcher intended to throw the ball (Ref. [A11]). A pitcher who aims for the upper-outer corner on an 0-2 count and misses by two inches off the plate is executing at an elite level, whereas a pitcher who aims down-and-away and misses by 14 inches into the heart of the zone has suffered a severe mechanical failure, even if Location+ scores the middle-middle strike favorably (Ref. [A11]). Empirical pitch densities for Zack Wheeler against right-handed batters illustrate this count-dependent targeting policy: in early and hitter-friendly counts, Wheeler concentrates fastballs around two primary targets (middle-middle and middle-away) safely inside the zone, whereas in pitcher-friendly counts he deliberately shifts his strategy and targets up-and-away on the zone edge where a potential ball is less costly (Ref. [A11]).

To decouple strategic intent from physical execution, the platform implements xCTRL via a contextual Gaussian Mixture Model (GMM) fitted with the Expectation-Maximization (EM) algorithm (Ref. [A11]). For a given pitcher i, pitch type m, batter handedness $h \in \{\text{L}, \text{R}\}$, and count state bin $c \in \{\text{early}, \text{hitter-friendly}, \text{pitcher-friendly}\}$, let $\mathcal{X}_{i,m,h,c} = \{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_N\}$ denote the set of two-dimensional ABS-normalized plate locations $\mathbf{u}_n = (\text{plate_x}_{\text{ABS},n}, \text{plate_z}_{\text{ABS}}$ converted to inches (Ref. [A2, A11]). The pitcher's location density is modeled as a K-component bivariate Gaussian mixture (Ref. [A11]):

$$p(\mathbf{u}_n \mid \Theta) = \sum_{k=1}^{K} \pi_k \, \mathcal{N}(\mathbf{u}_n \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \tag{7}$$

where π_k represents the prior probability of targeting intent cluster k ($\sum_{k=1}^{K} \pi_k = 1$), $\boldsymbol{\mu}_k \in \mathbb{R}^2$ represents the intended spatial target center of cluster k, and $\boldsymbol{\Sigma}_k \in \mathbb{R}^{2\times 2}$ is the execution covariance matrix (Ref. [A11]). During the E-step of the EM algorithm, the posterior responsibility γ_{nk} that pitch n was aimed at target $\boldsymbol{\mu}_k$ is computed as (Ref. [A11]):

$$\gamma_{nk} = \frac{\pi_k \, \mathcal{N}(\mathbf{u}_n \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{j=1}^{K} \pi_j \, \mathcal{N}(\mathbf{u}_n \mid \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)} \tag{8}$$

Once the EM algorithm converges (incorporating a smoothing-based extension across adjacent count bins when contextual sample sizes are limited), the pitch-level execution error xCTRL_n (in inches) is defined as the responsibility-weighted Euclidean distance between the realized pitch location $\mathbf{u}_n$ and the inferred target centroids $\boldsymbol{\mu}_k$ (Ref. [A11]):

$$\text{xCTRL}_n = \sum_{k=1}^{K} \gamma_{nk} \|\mathbf{u}_n - \boldsymbol{\mu}_k\|_2 \tag{9}$$

Across Major League Baseball, pitcher-level xCTRL exhibits a mean of 9.14 inches and a standard deviation of 1.10 inches, where lower values signify superior control (Ref. [A11]). Regressing Fielding Independent Pitching (FIP) simultaneously on Stuff+ and xCTRL proves that both metrics are highly significant and complementary: a one-standard-deviation improvement in Stuff+ (≈ 10 points) reduces expected FIP by ≈ 0.30 runs, which is identical in magnitude to the FIP reduction achieved by a 1.0-inch improvement in xCTRL (Ref. [A11]). At the granular pitch level, regressing RE24 on fastball velocity and fastball xCTRL demonstrates that control alone accounts for a difference of roughly 0.50 runs per start between the best and worst control pitchers over 20 batters faced ($0.0086 \times 3 \times 20 \approx 0.52$ runs), whereas each additional mile per hour of fastball velocity reduces runs allowed by ≈ 0.25 runs over the same 20-batter horizon (Ref. [A11]). For breaking balls, regressing ΔRE on curveball xCTRL while holding spin rate constant reveals that each 1.0-inch improvement in curveball xCTRL prevents 0.29 expected runs per 100 curveballs by eliminating poorly located middle-zone "hangers" (Ref. [A11]). Furthermore, kinematic tracking of collegiate pitchers using a TrackMan V3 unit (cohort mean fastball xCTRL $= 12.33 \pm 1.04$ in) demonstrates via backward multiple linear regression ($\alpha = 0.05$) that variability in torso angle at ball release explains 11.0% of the variance in fastball xCTRL ($F(1, 40) = 4.96, R^2 = 0.110, p = 0.032$), with every 0.332° increase in torso angle standard deviation increasing (degrading) xCTRL by 1.0 inch ($\beta = 0.332, t(40) = 2.23, p = 0.032$) (Ref. [A8]).

Pitch Metric	Model Specification	Distribution	Empirical Validation & Run Equivalence
Custom LightGBM Stuff+	Optuna-tuned LightGBM on pitch-level `xwOBA` / ΔRE24 with pitcher-grouped CV (Ref. [A10])	$\mu = 100.0$, $\sigma = 10.0$ (Ref. [A11])	Out-of-sample $r = +0.514$ with Whiff Rate, $r = +0.417$ with K Rate, and $r = -0.317$ with ERA (Ref. [A10]).
FanGraphs Stuff+ Benchmark	Proprietary pitch-quality baseline via `pybaseball.fg_pitching_data` (Ref. [A10, A13])	$\mu = 100.0$, $\sigma = 10.0$ (Ref. [A11])	Out-of-sample $r = +0.355$ with Whiff Rate (-0.159 vs. Custom), $r = +0.390$ with K Rate, $r = -0.358$ with ERA; 1 SD (+10) yields -0.30 FIP (Ref. [A10, A11]).
MLB xCTRL (Overall & FB)	Bivariate GMM fitted via EM over count/platoon bins (Ref. [A11])	$\mu = 9.14$ in, $\sigma = 1.10$ in (Ref. [A11])	-1.0 in in xCTRL yields -0.30 FIP (matching 1 SD Stuff+); ≈ 0.50 runs per 20 BF ($0.0086 \cdot 3 \cdot 20$) (Ref. [A11]).
MLB Curveball xCTRL	Bivariate GMM on breaking-ball intent clusters holding spin constant (Ref. [A11])	Pitch-level inches from $\boldsymbol{\mu}_k$ (Ref. [A11])	-1.0 in in curveball xCTRL reduces expected runs allowed by 0.29 runs per 100 curveballs (Ref. [A11]).
Torso Angle SD	Backward multiple linear regression on TrackMan V3 cohort (Ref. [A8])	Cohort xCTRL = 12.33 ± 1.04 in (Ref. [A8])	Explains 11.0% of FB xCTRL variance ($F(1, 40) = 4.96, R^2 = 0.110, p = 0.032$); $+0.332°$ SD degrades xCTRL by +1.0 in (Ref. [A8]).

1.4 Scalable Player Projection Architecture and Latent State-Space Engineering

Quantitative evaluations of contemporary baseball forecasting systems establish that baseline empirical regression architectures—such as Marcel or univariate state-space estimators—succeed in establishing overall population calibration but systematically fail to optimize individual player ordering (Ref. [B1, B7]). In controlled out-of-sample backtests spanning 704 major league hitter-seasons with 200 or more plate appearances across 2024 and 2025, an autoregressive state-space baseline achieved superior aggregate level calibration (a mean offset of +0.00106 wOBA versus +0.00691 for Steamer and +0.00497 for Dan Szymborski's ZiPS) and matched raw root mean square error (RMSE) benchmarks (.03121 versus .03086 for Steamer and .03006 for ZiPS) (Ref. [B2, B9, B16]). However, when level errors were recentered, both external commercial systems outperformed the baseline in player ordering by a statistically significant margin: ZiPS demonstrated a +4.71% skill advantage ($p = .003$) and Steamer achieved a +3.39% advantage ($p = .035$) (Ref. [B1, B13]).

Diagnostic profiling reveals the mathematical drivers of this ordering deficit: the omission of high-frequency Statcast contact-quality process metrics from the production filtering layer and the application of blunt, aggregate park adjustments rather than component-specific environmental shrinkage (Ref. [B1, B2]). Specifically, adjusting aggregate wOBA through a scalar batting park factor—while leaving underlying component rates unadjusted—induces a residual correlation of +0.0555 to +0.0616 with home ballparks, generating an unmitigated error slope of +0.00355 wOBA per 10 park factor points that distorts talent estimates in the tails of the stadium distribution (Ref. [B1, B2]). Furthermore, unconditioned minor-league transition models suffer from a −0.015 rookie selection bias when survival truncation is ignored, while traditional forecasting fails to capture the physical constraints governing pitcher role transitions (Ref. [B4]).

1.4.1 Quantitative Mechanics of Industry Forecasting Systems

Contemporary baseball projection systems operate on distinct statistical paradigms to separate persistent athletic skill from sample variance, apply aging trajectories, and translate minor league performance to MLB (Ref. [B1, B10, B11]):

- **Marcel the Monkey (Tom Tango):** Executes a deterministic multi-year weighted average of past performance, assigning weights of 5, 4, and 3 to seasons t, $t-1$, and $t-2$, respectively, and isolates true talent by applying a fixed regression constant (1,200 PA / 120 BF) toward the league mean alongside a deterministic piecewise linear aging factor (+0.6% / −0.3%) (Ref. [B7, B8, B15]).
- **Steamer (Cross, Davidson, & Rosenbloom):** Uses multilevel Generalized Linear Mixed Models (GLMM)

and Empirical Bayes shrinkage with parameter-specific maximum likelihood shrinkage across variable samples and non-parametric aging curves conditioned on athletic profiles (Ref. [B9, B11, B12, B14]).

- **ZiPS (Dan Szymborski):** Implements a composite architecture centered on comparable player matching (k-nearest neighbors) and internal "zStats" (such as zBABIP incorporating batted-ball spray distribution, Statcast sprint speed, and plate discipline) over 4-year weighted histories, replacing fixed cohort aging curves with dynamically calculated trajectories derived from historical player clusters (Ref. [B16, B17, B18, B19, B20, B21, B64]).
- **PECOTA (Nate Silver / Baseball Prospectus):** Employs Mahalanobis distance and historical k-NN over multi-year performance, height, weight, position, and minor league data to generate probabilistic percentile distributions from comparable player forward trajectories (Ref. [B22, B23, B24, B25, B26, B27, B28, B29]).
- **THE BAT / THE BAT X (Derek Carty):** Uses a machine learning ensemble and kinematic simulation combining past outcome rates with over 150 Statcast covariates (prioritizing 90th-percentile exit velocity, barrel percentages, sweet-spot frequencies, and sprint speeds) run through context-neutral physical simulations accounting for weather, temperature, humidity, wind vectors, and umpire tendencies (Ref. [B30, B31, B32, B33, B34, B35]).

System	Statistical Architecture	Regression & Shrinkage	Aging Formulation	Primary Inputs
Marcel (Ref. [B7, B8])	Heuristic Weighted Average (5/4/3)	Static population shrinkage (1,200 PA / 120 BF) toward league mean (Ref. [B7])	Deterministic piecewise linear (+0.6% / −0.3%) (Ref. [B8])	3-year box-score history, basic age (Ref. [B7])
Steamer (Ref. [B9, B11])	Multilevel GLMM / Empirical Bayes	Parameter-specific ML shrinkage across variable samples (Ref. [B12])	Non-parametric aging conditioned on athletic profiles (Ref. [B14])	Pitch tracking, plate discipline, box scores (Ref. [B12])
ZiPS (Ref. [B16, B17])	Composite Time-Series & k-NN Analogs	Multi-year baseline adjusted via internal zStats (zBABIP) (Ref. [B18])	Dynamic aging via cluster-matched historical analogs (Ref. [B16])	4-yr box scores, spray angles, sprint speed (Ref. [B18])
PECOTA (Ref. [B22, B23])	Mahalanobis Distance / Historical k-NN	Empirical distribution matching over nearest cohorts (Ref. [B25])	Probabilistic distribution from comparable trajectories (Ref. [B26])	Multi-yr stats, height, weight, pos, MiLB (Ref. [B25])
THE BAT/X (Ref. [B30, B31])	ML Ensemble / Kinematic Simulation	Multi-layer regression with Statcast process expectations (Ref. [B31])	Trajectory decay conditioned on physical traits & EV decline (Ref. [B33])	Statcast EV/LA, park geometry, microclimate (Ref. [B31])

1.4.2 Contact Quality Stabilization and Batted-Ball Distributional Modeling

The primary empirical vulnerability of a pure box-score state-space model is the treatment of Batting Average on Balls in Play (BABIP), which requires 820 Balls in Play (BIP) to stabilize and exhibits a negative rank correlation (-0.07) for rookie forecasts derived purely from minor league box scores (Ref. [B1, B2, B36, B37]). Incorporating fast-stabilizing Statcast batted-ball process metrics and launch angle tightness directly remedies this vulnerability (Ref. [B1, B37, B38, B39, B41, B42, B43, B44, B45]):

Metric Category	Statistical Metric	Stabilization Threshold	Mechanistic Significance in Projection Layer
Batted-Ball Process	Exit Velocity (Mean EV) (Ref. [B37])	50 Balls in Play (BIP)	Direct measure of kinetic energy transfer; highly stable indicator of raw power ceiling (Ref. [B37, B38, B39]).

Metric Category	Statistical Metric	Stabilization Threshold	Mechanistic Significance in Projection Layer
Batted-Ball Process	90th Percentile Exit Velo (EV90) (Ref. [B1])	60 Balls in Play (BIP)	Eliminates drag of mis-hit balls (topped/flared); isolates high-end authority independent of pitch type (Ref. [B1]).
Contact Authority	Barrel Rate (BRL%) (Ref. [B37])	50 Balls in Play (BIP)	Captures optimal launch angle and velocity window; drives isolated power (Ref. [B37, B38]).
Contact Authority	Hard-Hit Rate (Hard-Hit%) (Ref. [B37])	50 Balls in Play (BIP)	Percentage of batted balls $\geq$ 95 mph; robust proxy for baseline contact authority (Ref. [B37, B41]).
Plate Discipline	Whiff Rate (Whiff%) (Ref. [B1])	60 Swings	Swings missing ball; fast-stabilizing leading indicator for true strikeout trajectory (Ref. [B1]).
Plate Discipline	Chase Rate (O-Swing%) (Ref. [B1])	100 Pitches Out-of-Zone	Propensity to expand strike zone; directly conditions forward walk-rate expectations (Ref. [B1]).
Outcome Channel	Raw BABIP (Ref. [B37])	820 Balls in Play (BIP)	Dominated by defensive positioning, park area, and binomial luck; requires 3+ years of data (Ref. [B37]).

1.4.3 Hierarchical Empirical Bayes Estimation of Component Ballpark Factors

To eliminate the +0.00355 wOBA per 10 BPF points error slope (95% clustered CI) across extreme environments like Coors Field (129 BPF) or T-Mobile Park (88 BPF) (Ref. [B1, B2]), the model formulates component park effects through a Bayesian hierarchical linear mixed model with partial pooling (Ref. [B2, B8, B46, B47, B48, B49, B50, B51, B52, B65, B66, B67, B68]):

$$y_{c,t,j,k} = \mu_c + \alpha_{c,j} + \beta_{c,k} + \gamma_{c,j,h} + \epsilon_{c,t,j,k}, \quad \alpha_{c,j} \sim \mathcal{N}(\mathbf{X}_j \boldsymbol{\gamma}_c, \tau_c^2) \tag{10}$$

where $y_{c,t,j,k}$ represents the log-rate of component c in season t at venue j against opponent k, $\alpha_{c,j}$ is the persistent stadium effect shrunken toward physical venue covariates $\mathbf{X}_j \boldsymbol{\gamma}_c$ (elevation, temperature-adjusted air density, foul pole distances, wall height, and foul territory square footage), $\beta_{c,k}$ is the opponent quality adjustment, and $\gamma_{c,j,h}$ is the venue-by-batter-handedness interaction (Ref. [B2, B46]).

Component	Physical Stadium Drivers	1-Yr Rel. (r)	EB Shrinkage Target ($\mathbf{X}_j\boldsymbol{\gamma}$)	MLB Factor Range
Home Runs (HR) (Ref. [B2, B45])	Fence distance, wall height, elevation, air density, humidor (Ref. [B2])	High	Aerodynamic trajectory simulation over 3D stadium scan	70 (Oracle Park) to 135 (Great American) (Ref. [B46])
BABIP (Ref. [B2, B45])	Outfield acreage, turf resistance, foul territory size (Ref. [B2])	Moderate	Playable fair surface area minus defensive coverage	92 (Fenway Park) to 118 (Coors Field) (Ref. [B2])
Extra-Base Hits (2B/3B) (Ref. [B2])	Corner angles, alley depth, carom reflectivity, wall height (Ref. [B2])	Moderate	Square footage of outfield gaps and wall height	80 (Yankee Stadium) to 130 (Fenway Park) (Ref. [B2])
Strikeouts (K) (Ref. [B2, B47])	Batter-eye contrast, stadium light lux, shadows (Ref. [B2])	Very Low	League mean prior (100), zero dimensional loading	96 to 104 (Tight convergence around 100) (Ref. [B2])
Walks (BB) (Ref. [B2, B47])	Umpire visual perspective, background noise behind plate (Ref. [B2])	Very Low	League mean prior (100), zero dimensional loading	97 to 103 (Tight convergence around 100) (Ref. [B2])

1.4.4 Minor League Translation Matrices and Multivariate Dynamic Linear Models (VectorSSM)

Forecasting rookie performance builds on Clay Davenport's Translations (DTs), Bill James' Major League Equivalencies (MLEs), and Chris Mitchell's KATOH probit regressions conditioned on age relative to level, plate discipline, isolated power, and position, while correcting for the -0.015 rookie selection bias via survival-truncation risk-set weighting across MiLB `sportIds` 11 (AAA), 12 (AA), 13 (High-A), and 14 (Single-A) linked via the Chadwick Bureau Register (Ref. [B3, B4, B6, B40, B53, B54, B55, B56, B57, B58]).

At the mathematical core of the projection engine is an autoregressive multivariate state-space model (**VectorSSM**), replacing independent scalar filters (Ref. [B5, B59]). For each player and season t, observation variance is modeled via heteroskedastic binomial variance evaluated at the player's own predicted rate $\pi_{j,t}$ (Ref. [B1, B5]):

$$H_{t,jj} = \frac{\pi_{j,t}(1-\pi_{j,t})}{N_{j,t}} \tag{11}$$

The Kalman filter equations execute recursive Bayesian updates sequentially across each player's career history (Ref. [B5, B59]):

$$\textbf{One-Step State \& Covariance Prediction:} \quad \mathbf{a}_{t|t-1} = \mathbf{T}_t\mathbf{a}_{t-1} + \mathbf{c}_t, \quad \mathbf{P}_{t|t-1} = \mathbf{T}_t\mathbf{P}_{t-1}\mathbf{T}_t^T + \mathbf{Q}_t \tag{12}$$

$$\textbf{Innovation \& Residual Covariance:} \quad \mathbf{v}_t = \mathbf{y}_t - (\mathbf{Z}_t\mathbf{a}_{t|t-1} + \mathbf{d}_t), \quad \mathbf{F}_t = \mathbf{Z}_t\mathbf{P}_{t|t-1}\mathbf{Z}_t^T + \mathbf{H}_t \tag{13}$$

$$\textbf{Kalman Gain \& State Update:} \quad \mathbf{K}_t = \mathbf{P}_{t|t-1}\mathbf{Z}_t^T\mathbf{F}_t^{-1}, \quad \mathbf{a}_t = \mathbf{a}_{t|t-1} + \mathbf{K}_t\mathbf{v}_t, \quad \mathbf{P}_t = (\mathbf{I} - \mathbf{K}_t\mathbf{Z}_t)\mathbf{P}_{t|t-1} \tag{14}$$

For multi-horizon forecasting ($h \geq 1$), the state vector and state error covariance are propagated forward recursively (Ref. [B5]):

$$\mathbf{a}_{t+h|t} = \left(\prod_{k=1}^{h}\mathbf{T}_{t+k}\right)\mathbf{a}_t + \sum_{k=1}^{h}\left(\prod_{j=k+1}^{h}\mathbf{T}_{t+j}\right)\mathbf{c}_{t+k}, \quad \mathbf{P}_{t+h|t} = \mathbf{T}_{t+h}\mathbf{P}_{t+h-1|t}\mathbf{T}_{t+h}^T + \mathbf{Q}_{t+h} \tag{15}$$

Given linear wOBA weights $\mathbf{w} \in \mathbb{R}^m$, the projected mean and exact predictive variance are propagated directly without scalar delta-method approximations (Ref. [B5]):

$$\mathbb{E}[\text{wOBA}_{t+h}] = \mathbf{w}^T\mathbf{a}_{t+h|t} + \mu_{\text{wOBA}}, \quad \text{Var}(\text{wOBA}_{t+h}) = \mathbf{w}^T\mathbf{P}_{t+h|t}\mathbf{w} + \sigma^2_{\text{rebuild}} \tag{16}$$

Internal backtesting of VectorSSM demonstrated univariate rebuild parity (RMSE .0314) and verified nominal coverage across forward horizons ($h = 1$ to $h = 3$ coverage bounded between .779 and .793 for nominal 80% intervals) while eliminating 15 free parameters across the joint estimation space, capturing the strong positive drift correlation ($r = +0.615$) between latent home run power ($HR\%$) and non-HR extra-base hit power ($XBH\%$) while confirming that BABIP drift remains orthogonal to power components (Ref. [B5]).

For comparable player matching, the Mahalanobis distance function $D_M(\mathbf{x}, \mathbf{y}_i)$ is scaled using a localized adaptive Gaussian kernel where σ_{local} is the distance to the k-th nearest neighbor ($k = 30$), maintaining effective sample sizes within the optimal calibration band of 20 to 45 players (Ref. [B7, B60]):

$$w_i = \exp\left(-\frac{D_M(\mathbf{x}, \mathbf{y}_i)^2}{2\sigma^2_{\text{local}}}\right) \tag{17}$$

1.4.5 MLB Surplus Valuation and Implementation Validation Tiers

When translating Stuff+, xCTRL, and VectorSSM projections into front-office contract valuations (allocating a fixed pool of 570 replacement runs for position players and role-split replacement levels for pitchers) (Ref. [B4, B6, B60]), the platform anchors its surplus value converter to the 2025–2026 MLB free-agent market equilibrium (Ref. [A14]). While position players and starting pitchers projected for 2 + WAR command an average of \$7.7M/WAR (ranging from \$7.28M/WAR for Tier-1 All-Stars like Framber Valdez, Dylan Cease, Bo Bichette, and Alex Bregman and \$7.62M/WAR for Tier-2 All-Stars to \$7.95M/WAR for Kyle Tucker and \$10.3M/WAR for Pete Alonso), high-leverage relievers (Edwin Diaz, Devin Williams, Robert Suarez) command \$17.5M/WAR (Ref. [A14]). Thus, bullpen arms are valued on a Dollar-per-Win-Probability-Added (\$/WPA) and convex tier pricing curve (Ref. [A14, B4]).

Tier & Preconditions	Technical Implementation Tasks	Validation Gate & Threshold	Failure Modes & Mitigations
Tier 0: Ingestion Spine (Ref. [B40])	Ingest MiLB box scores (2005–2020) via MLB Stats API (`sportIds` 11–14); FanGraphs Guts & Savant FRV; join Chadwick ID register (Ref. [B3, B40]).	Match rate $\geq$ 99.5% on Chadwick keys; wOBA identity closure; zero missing season weights (Ref. [B7]).	*Failure:* Polars OOM on pitch logs; API throttling. *Mitigation:* Lazy parquet scans; 2 req/sec backoff (Ref. [B7, B40]).
Tier 1: Core Statistical Remediation (P0) (Ref. [B1, B2])	Implement hierarchical EB park factors (HR, BABIP, XBH, K, BB); wire EV90, BRL% into hitter matrix; deploy age-relative MiLB prior (Ref. [B1, B2, B6]).	Park residual ≈ 0; wOBA error slope eliminated; hitter ordering skill gain +3.4% to +4.7% (Ref. [B1, B2, B5]).	*Failure:* Collinear tracking singularity. *Mitigation:* Ridge penalty on observation matrix; 50 BIP floor (Ref. [B7]).
Tier 2: VectorSSM & Kinematics (Ref. [B4, B5])	Deploy VectorSSM with joint drift covariance $\mathbf{Q}$; wire in-house Stuff+ & velocity into pitcher states; enforce SP $\rightarrow$ RP constraints (Ref. [B4, B5]).	Multi-horizon coverage bounded at .779–.793 (80% CI); pitcher K% correlation stabilized (Ref. [B4, B5]).	*Failure:* $\mathbf{Q}$ losing PSD during optimization. *Mitigation:* Parameterize $\mathbf{Q} = \mathbf{L}\mathbf{L}^T$ via Cholesky factor (Ref. [B7]).
Tier 3: Latent Comps & Asset Valuation (Ref. [B4, B60])	Rescale Mahalanobis comps via adaptive Gaussian kernel ($k = 30$); role-split pitcher replacement levels; convex tier contract pricing (Ref. [B4, B7, B60]).	Effective sample size 20–45; pitcher RAR closes within 0.1; valuation LOYO slope calibrated (Ref. [B4, B60]).	*Failure:* Kernel collapsing to uniform weights. *Mitigation:* Scale penalty by empirical distance SD; binned PAVA (Ref. [B4]).

1.5 2027 SABR Analytics Conference Research Proposals

In accordance with the Society for American Baseball Research (SABR) guidelines for the 2027 SABR Analytics Conference (abstract submission deadline: November 25, 2026; Sarah Langs Women in Baseball Scholarship and Research Awards deadline: December 18, 2026) (Ref. [A15]) and the blind-review standards established for SABR 54 (Ref. [A16]), the platform anchors its baseball vertical around two complementary research specifications:

1. **Abstract 1 — Decomposing Pitch-to-Swing Geometric Collisions Under the 2026 ABS Regime: Integrating Arm-Slot Release Kinematics, xCTRL Intent Clusters, and 3D Bat-Path Collision Geometry Deviation:** Utilizing 2.8 million Statcast pitches from 2024–2026 ingested via `pybaseball` and `python-mlb-statsapi` (Ref. [A18, A19]), all pre-2026 pitch trajectories are forward-projected from $y = 1.4167$ ft to the 2026 ABS middle-of-plate plane ($y = 0.7083$ ft) using 9-parameter kinematics (Ref. [A1, A2]). An EM Gaussian Mixture Model computes pitch-level intent centroids $\boldsymbol{\mu}_k$ and signed miss vectors $\mathbf{e}_n = \mathbf{u}_n - \boldsymbol{\mu}_k$ across count and platoon states (Ref. [A11]), while a continuous 3D Collision Geometry Deviation (CGD) tensor across 25 normalized ABS strike-zone cells models the angular discrepancy between the incoming pitch approach vector at $y = 0.7083$ ft (conditioned on pitcher `arm_angle` and induced break) and the batter's 40-ms pre-impact swing plane (`swing_path_tilt`, `attack_angle`, `attack_direction`, and `intercept_ball_minus_batter_pos_x/y_inches`), building on two-strike swing adjustment research (0 to +0.020 xwOBA, up to +0.038 for Yordan Alvarez) (Ref. [A2, A8, A17]). Preliminary modeling indicates that unintended miss vectors ($\|\mathbf{e}_n\|_2 > 11$ in) crossing opposing seam/tunnel planes inflate batter contact depth error by 2.4 inches, whereas misses drifting toward the batter's natural `swing_path_tilt` plane suffer a 42% surge in barrel rate (Ref. [A2, A8, A11]).

2. **Abstract 2 — Dynamic Arsenal Sequencing under Command Decay: An `xLSTM` and Causal Collision Geometry Framework for Live Bullpen Optimization:** Combines within-game **Pitch Decay** and **Blend+ / ArsenalFX** repertoire interaction modeling with `xCTRL` cluster widening after pitch 60 and an `xLSTM` neural network (`pitchpredict`) ingesting live MLB `/feed/live/diffPatch` streams to output optimal next-pitch sequences and automated bullpen reliever alerts backed by SHAP feature attributions and Traceable Hybrid RAG provenance (Ref. [A8, A10, A11, A84]).

2 Phase 2: National Basketball Association — Possessional RAPM, 2026–27 CBA Apron Mechanics, and Trade Liquidity (ACRE)

2.1 Play-by-Play Lineup Reconstruction and Scorekeeper-Unbiased Possession State Machine

Because the NBA does not expose an error-free public lineup stint feed for recent seasons, the basketball ingestion pipeline reconstructs every 10-man floor unit from raw play-by-play event streams (`playbyplayv3` via `nba_api` and `hoopR-nba-data` via SportsDataverse) cross-referenced against game box scores, validating player stint intervals to within < 0.5 seconds per game (Ref. [A18, A20, A21]).

Legacy basketball pipelines estimate possessions from aggregate box scores via the standard Dean Oliver estimator (Ref. [A22]):

$$\widehat{\text{POSS}}_{\text{home}} = \text{FGA}_{\text{home}} - \text{OREB}_{\text{home}} + (\text{TO}_{\text{home}} + \varepsilon_{\text{TO}}) + 0.44 \cdot \text{FTA}_{\text{home}} \tag{18}$$

Data audits demonstrate that scorekeeper bias—specifically the inflation of home steals, which forces a corresponding phantom turnover ($\varepsilon_{\text{TO}} > 0$) onto the visiting team's ledger—propagates upward bias into box-score possession estimates for both teams (Ref. [A22]). To eliminate scorekeeper bias and stint-boundary apportionment distortion, the platform implements a direct play-by-play possession state machine where a possession terminates only upon a made field goal, defensive rebound, turnover, or final made free throw, collapsing multi-FTA foul sequences into a single possession, excluding technical foul free throws from possession counts, and apportioning fractional possession weights across mid-possession substitution stint boundaries (Ref. [A22, A23]).

2.2 Regularized Adjusted Plus-Minus (RAPM), Box-Prior IPM, and Extended E-RAPM

Let P denote the total number of active NBA players and N denote the total number of unique 10-man stints ($\approx 30,000$–$60,000$ stints over a multi-season window) (Ref. [A22, A24]). The sparse design matrix $X \in \mathbb{R}^{N \times (2P+1)}$ sets column 0 to 1.0 (baseline efficiency), columns $1, \ldots, P$ to $+1$ for offensive players on court (O-RAPM), and columns $P+1, \ldots, 2P$ to -1 for defensive players on court (D-RAPM, so positive β_{P+k} signifies points prevented per 100 possessions) (Ref. [A21, A23, A25]). Because $\text{rank}(X^T W X) \leq 2P - 1 < 2P + 1$, Ordinary Least Squares is strictly singular; applying Tikhonov (ℓ_2) regularization with possession weights $W = \text{diag}(w_1, \ldots, w_N)$ and penalty $\lambda \approx 2900$ yields the weighted RAPM estimator (Ref. [A22, A23, A24, A25, A26]):

$$\hat{\boldsymbol{\beta}}_{\text{RAPM}} = (X^T W X + \lambda I)^{-1} X^T W \mathbf{y} \tag{19}$$

To resolve four documented flaws in public RAPM implementations, the platform incorporates:

1. **Dummy RAPM:** Collapses all players below 250 possessions into a composite replacement-level offensive and defensive dummy column rather than deleting low-minute columns, preserving exact 5-on-5 row sums (Ref. [A25, A27]).
2. **Box-Prior Impact Plus-Minus (IPM):** Constructs an independent prior vector $\boldsymbol{\beta}_0 \in \mathbb{R}^{2P+1}$ using Ben Taylor's Box Creation (incorporating a volume-gated 3-point proficiency sigmoid in 3PA × 3P% floored at zero), Offensive Load, and Shot-Zone-Adjusted eFG% Residual, solving ridge regression on the prior-adjusted residuals $\mathbf{y}^* = \mathbf{y} - X\boldsymbol{\beta}_0$ (Ref. [A21, A28]):

$$\hat{\boldsymbol{\beta}}_{\text{IPM}} = \boldsymbol{\beta}_0 + (X^T W X + \lambda I)^{-1} X^T W (\mathbf{y} - X\boldsymbol{\beta}_0) \tag{20}$$

3. **Full Posterior Covariance:** Reports player evaluations as posterior probability distributions $\mathcal{N}(\hat{\beta}_i, \Sigma_{\text{post},ii})$ using $\boldsymbol{\Sigma}_{\text{post}} = \hat{\sigma}^2 (X^T W X + \lambda I)^{-1}$ and possession-weighted offensive/defensive totals (Ref. [A22, A25]).
4. **Extended RAPM (E-RAPM):** Augments X with combinatorial synergy indicator columns Z generated via $C_{5,x} = \binom{5}{x} = \frac{5!}{x!(5-x)!}$ for $x \in \{2, 3, 4, 5\}$ (10 pairs, 10 trios, 5 quartets, 1 five-man unit per stint) with ridge penalty $\lambda_{\text{syn}} > \lambda$ to isolate non-additive lineup synergy (Ref. [A21, A27]).

2.3 2026–27 NBA CBA 1st and 2nd Apron Algorithmic Rules Engine

Under the 2023 NBA–NBPA Collective Bargaining Agreement (CBA, Article VII) for the 2026–27 season (October 19, 2026 through June 19, 2027; trade deadline February 11, 2027), the Salary Cap is \$164,961,000 (\$165.0M),

Minimum Team Salary Floor (90%) is \$148.5M, Luxury Tax Threshold (121.5%) is \$200.4M, First Apron is \$209,015,000 (\$209.0M), Second Apron is \$221,686,000 (\$221.7M), and the cap-indexed middle-tier trade matching constant ("\$7.5M" in 2023–24) scales under Art. VII §6(j)(1)(iv) to exactly \$9,095,709 (\$9.1M) (Ref. [A29, A30, A31, A32]). The deterministic constraint solver enforces maximum incoming salary $S_{\text{in, max}}$ given outgoing salary $S_{\text{out}} = \sum_{j \in \mathcal{O}} s_j$ and post-transaction payroll P_{post} (Ref. [A29, A32, A33]):

$$S_{\text{in, max}}(S_{\text{out}}, P_{\text{post}}) = \begin{cases} (164{,}961{,}000 - P_{\text{pre}}) + S_{\text{out}} + 100{,}000 & \text{if } P_{\text{post}} < 164{,}961{,}000 \text{ (Under Cap Room)} \\ 2.00 \cdot S_{\text{out}} + 250{,}000 & \text{if } P_{\text{post}} < 209{,}015{,}000 \text{ and } S_{\text{out}} \le 9{,}095{,}709 \\ S_{\text{out}} + 9{,}095{,}709 & \text{if } P_{\text{post}} < 209{,}015{,}000 \text{ and } 9{,}095{,}709 < S_{\text{out}} \le 35{,}200{,}000 \\ 1.25 \cdot S_{\text{out}} + 250{,}000 & \text{if } P_{\text{post}} < 209{,}015{,}000 \text{ and } S_{\text{out}} > 35{,}200{,}000 \\ 1.00 \cdot S_{\text{out}} & \text{if } 209{,}015{,}000 \le P_{\text{post}} < 221{,}686{,}000 \text{ (First Apron)} \\ 1.00 \cdot S_{\text{out}} \text{ subject to } |\mathcal{O}| = 1 & \text{if } P_{\text{post}} \ge 221{,}686{,}000 \text{ (Second Apron: No Aggregation)} \end{cases} \tag{21}$$

2026–27 Payroll Tier	Statutory Level	Max Incoming ($S_{\text{in, max}}$)	Exception & S&T Access	Draft, Cash & Hard-Cap Triggers
Under Salary Cap (Floor: \$148.5M)	< \$164,961,000 (Ref. [A29, A32])	Cap Room + \$100K cushion (Ref. [A32])	Room MLE (\$9.4M, 3 yrs); forfeits Non-Tax MLE & BAE (Ref. [A31, A32])	No hard cap unless S&T acquired (Ref. [A32])
Over Cap / Below 1st Apron (Tax: \$200.4M)	\$164.961M to \$209.015M (Ref. [A29, A30, A32])	Tier 1 (≤ \$9.1M): 200% + \$250K; Tier 2 (\$9.1M–\$35.2M): +\$9,095,709; Tier 3 (> \$35.2M): 125% + \$250K (Ref. [A29, A32])	Non-Taxpayer MLE (\$15.0M, 4 yrs), BAE (\$5.5M, 2 yrs), TPE +\$250K (Ref. [A32])	Expanded matching ($S_{\text{in}} > S_{\text{out}}$), Non-Tax MLE, BAE, or S&T **hard-caps at 1st Apron (\$209.015M)** (Ref. [A29, A32])
First Apron Team	\$209,015,000 to \$221,686,000 (Ref. [A30, A32])	100% of outgoing (dollar-for-dollar) (Ref. [A29, A32])	Taxpayer MLE (\$6.1M); no prior-yr TPEs; no incoming S&T; buyout ban >MLE (Ref. [A31, A32, A34])	Taxpayer MLE, aggregating salaries, or sending cash **hard-caps at 2nd Apron (\$221.686M)** (Ref. [A32, A34])
Second Apron Team	≥ \$221,686,000 (Ref. [A30, A32])	100% of single outgoing contract (**zero aggregation**) (Ref. [A29, A32])	Zero MLE access; zero TPE access (current or prior year) (Ref. [A31, A32, A34])	Zero cash out; **1st-rd pick 7 yrs out (2034) frozen**; drops to end of 1st rd if ≥2nd Apron in ≥ 2 of next 4 yrs (Ref. [A31, A32, A34])

2.4 Trade Liquidity Index (TLI) and Apron-Constrained Roster Efficiency (ACRE)

Modeling the 30 NBA rosters as a directed hypergraph $\mathcal{G}_{\text{trade}} = (\mathcal{V}, \mathcal{E})$ where hyperedges $e \in \mathcal{E}$ represent all valid 2-team and 3-team exchanges satisfying Art. VII §6 (including multi-player sub-transaction decomposability for Second Apron teams and Art. VII §8 2-month re-aggregation locks) (Ref. [A29, A32, A33]), the Player Trade Liquidity Index ($\text{TLI}_i \in [0,1]$) and Apron-Constrained Roster Efficiency (ACRE_i) are formulated as (Ref. [A21, A27, A31, A32, A35]):

$$\text{TLI}_i(T) = \frac{1}{\sum_{T' \neq T} \sum_{j \in T'} V_j} \sum_{T' \neq T} \sum_{\substack{j \in T' \\ \exists e \in \mathcal{E} : i \in \mathcal{O}_T, j \in \mathcal{I}_T}} V_j \tag{22}$$

$$\text{ACRE}_i = \frac{\Delta\text{Wins}_i(\hat{\beta}_{\text{IPM},i}, \hat{\gamma}_{\text{syn}})}{c_i + \Delta\text{Tax}(c_i, P_T) + \lambda_{\text{liq}}(1 - \text{TLI}_i(T)) \cdot c_i + \mathbb{I}(P_T \ge 221.686\text{M}) \cdot \mathbb{E}[\Delta V_{\text{pick},t+7}]} \tag{23}$$

3 Phase 3: National Hockey League — Optical Edge Tracking, Expected Goals (xG/GSAx), and fastRhockey Integration

3.1 Direct Ingestion of `api-web.nhle.com/v1` and NHL Edge Optical Telemetry

Following the deprecation of `statsapi.web.nhl.com`, the hockey pipeline integrates `https://api-web.nhle.com/v1` and `https://api.nhle.com/stats/rest` via `fastRhockey` v1.0.0 (released August 25, 2026) and `sportsdataverse-py` (`fastRhockey-nhl-data`), providing 22 PWHL women's hockey functions and 33 NHL Edge optical tracking wrappers (Ref. [A20, A36, A37, A38, A39, A40, A41, A42, A43, A44, A45, A46]). Because seasonal roster stubs (`/v1/roster/{team}/{season}`) occasionally omit active players (e.g., Oliver Ekman-Larsson on FLA 2023–24 or Kyle Burroughs on SJS 2023–24), master rosters are constructed from the union of `/v1/gamecenter/{gameId}/boxscore` and shift-chart player IDs hydrated via `/v1/player/{playerId}/landing` (Ref. [A36, A38, A44, A47]).

Data Domain	Direct `api-web.nhle.com/v1` Path	`fastRhockey` v1.0.0	Extracted Telemetry
Game PBP & Shifts	`/v1/gamecenter/{gameId}/play-by-play`, `/boxscore` (Ref. [A36, A38])	`nhl_game_feed()`, `load_nhl_pbp()` (Ref. [A39, A43])	Coordinates (x, y), shooter/goalie IDs, shot types, on-ice shift intervals (Ref. [A38, A43]).
Skater Optical (Edge)	`/v1/edge/skater-shot-speed-detail/{id}/{season}/{type}`, `skater-shot-location-detail` (Ref. [A44])	`nhl_edge_skater_detail()` (33 Edge funcs) (Ref. [A39, A42, A43])	Top shot speed, 90th-% shot speed, bursts $> 20/22$ mph, skating distance (Ref. [A43, A44, A45]).
Goalie Optical (Edge)	`/v1/edge/goalie-shot-location-detail/{id}/{season}/{type}`, `goalie-comparison/{id}/now` (Ref. [A44])	`nhl_edge_goalie_save_percentage_detail()` (Ref. [A43])	High/mid/low danger spatial save %, crease depth & recovery metrics (Ref. [A43, A44]).
Team Optical (Edge)	`/v1/edge/team-skating-speed-detail/{id}/{season}/{type}`, `team-shot-speed-detail` (Ref. [A44, A45])	`nhl_edge_team_detail()`, `..._shot_speed_detail()` (Ref. [A45, A46])	Team skating distance, OZ possession time percentiles, velocity distributions (Ref. [A39, A44, A45]).
PWHL Women's Pro	HockeyTech / PWHL API feeds (Ref. [A20, A39])	`pwhl_pbp()`, `load_pwhl_pbp()` (Ref. [A39])	Full PBP, skater/goalie boxscores, transactions, and xG parity (Ref. [A39, A43]).

3.2 Expected Goals (xG), Goaltender GSAx, and 4-State Special Teams RAPM

Validated across 117,179 clean shot attempts per season (after pruning 3,119 incomplete records from 120,298 rows and removing behind-the-net shots $|x| > 89$ ft), the feature pipeline transposes `arenaAdjustedXCord` and `arenaAdjustedYCord` onto $+x$ ($x_{\text{std}} = |x_{\text{adj}}|$, $y_{\text{std}} = \text{sign}(x_{\text{adj}}) \cdot y_{\text{adj}}$), computes `timeSinceLastEvent` ($\Delta t$) and `speedFromLastEvent` ($v_{\text{prior}} = \|\mathbf{p}_t - \mathbf{p}_{t-1}\|_2 / \Delta t$), applies one-hot encoding to `shotType`, and joins NHL Edge optical shot velocity with a hierarchical Bayesian shooter finishing random effect $\alpha_{\text{shooter}} \sim \mathcal{N}(0, \sigma^2_{\text{fin}})$ (Ref. [A39, A44, A50, A51, A52, A53, A54, A55, A56]). Gradient-boosted trees (`XGBoost`, tree depth 3–6, $\eta = 0.1$) achieve validation ROC-AUC scores of 0.78 to 0.82, and shot probabilities are aggregated into Goaltender Goals Saved Above Expected (Ref. [A48, A49, A50, A51]):

$$\text{GSAx}_g = \sum_{i \in \mathcal{S}_g} \widehat{\text{xG}}_i - \sum_{i \in \mathcal{S}_g} \text{Goal}_i \tag{24}$$

Skater on-ice impact is decomposed via weighted ridge regression (`RAPM_CF` and `RAPM_xG`) into Even-Strength and four Special Teams predictor states: Powerplay Offense, Powerplay Defense, Shorthanded Offense, and Shorthanded Defense (Ref. [A53]).

4 Phase 4: National Football League — Spatio-Temporal Tracking, Disguise Entropy, and Draft Surplus Valuation

4.1 Big Data Bowl Coordinate Standardization and Continuous Spatio-Temporal EPA (st-EPA)

Ingesting play-by-play and Next Gen Stats via `nfl_data_py` v0.3.3 (`downcast=True` reducing memory by $\sim 30\%$) and 10-Hz NFL Big Data Bowl tracking data (Ref. [A20, A57, A58, A59, A60, A61]), all left-moving plays (playDirection == 'left') across the 120×53.333 yd surface are reflected to advance left-to-right ($+x$) (Ref. [A58]):

$$x_{\text{std}} = 120 - x, \quad y_{\text{std}} = \frac{160}{3} - y, \quad \theta_{\text{dir, std}} = (\theta_{\text{dir}} + 180^\circ) \bmod 360^\circ, \quad \theta_{\text{o, std}} = (\theta_{\text{o}} + 180^\circ) \bmod 360^\circ \tag{25}$$

$$v_x = s \cdot \sin\left(\frac{\pi}{180}\theta_{\text{dir, std}}\right), \quad v_y = s \cdot \cos\left(\frac{\pi}{180}\theta_{\text{dir, std}}\right) \tag{26}$$

Using a frame-level graph attention network over the 22-player state tensor $\mathbf{S}_t \in \mathbb{R}^{22\times 10}$ at every 100 ms frame t, Continuous Spatio-Temporal Expected Points (st-EP$_t$) integrates target probability $p_t(r \mid \mathbf{S}_t)$, completion probability $\widehat{\text{CP}}_{t,r}(\mathbf{S}_t)$, and expected YAC density $f_{t,r}(y \mid \mathbf{S}_t)$ (Ref. [A57, A59, A61]):

$$\text{st-EP}_t = \sum_{r=1}^{5} p_t(r \mid \mathbf{S}_t) \left[\widehat{\text{CP}}_{t,r}(\mathbf{S}_t) \int_{\mathcal{Y}} E_0(\text{next_state}(y)) f_{t,r}(y \mid \mathbf{S}_t)\, dy + (1 - \widehat{\text{CP}}_{t,r}(\mathbf{S}_t)) E_0(\text{inc})\right], \quad \text{st-EPA}_t = \text{st-EP}_t - \text{EP}_{t_{\text{snap}}} \tag{27}$$

This partitions every passing play into Route/Protection Creation ($\Delta\text{st-EPA}_{\text{scheme}} = \text{st-EP}_{t_{\text{throw}}} - \text{EP}_{t_{\text{snap}}}$), Ball-in-Air Value ($\Delta\text{st-EPA}_{\text{air}} = \text{st-EP}_{t_{\text{arrival}}} - \text{st-EP}_{t_{\text{throw}}}$), and YAC/Tackling Execution ($\Delta\text{st-EPA}_{\text{YAC}} = \text{EP}_{\text{final}} - \text{st-EP}_{t_{\text{arrival}}}$).

To quantify defensive disguise across the $K = 7$ canonical coverage shells $\mathcal{C} = \{\text{C0}, \text{C1}, \text{C2}, \text{C2-Man}, \text{C3}, \text{C4}, \text{C6}\}$, the pipeline computes Pre-Snap Shannon Disguise Entropy (H_{snap}) at t_{snap} and Pre-to-Post-Snap Kullback-Leibler Deception Shock (D_{KL}) between $t_0 = t_{\text{snap}} - 0.5$ s and $t_1 = t_{\text{snap}} + 1.5$ s:

$$H_{\text{snap}}(\mathcal{C} \mid \mathbf{S}_{t_{\text{snap}}}) = -\sum_{k=1}^{7} p(c_k \mid \mathbf{S}_{t_{\text{snap}}}) \log_2 p(c_k \mid \mathbf{S}_{t_{\text{snap}}}), \quad D_{\text{KL}}\left(P(\mathcal{C} \mid \mathbf{S}_{t_1}) \,\|\, P(\mathcal{C} \mid \mathbf{S}_{t_0})\right) = \sum_{k=1}^{7} p(c_k \mid \mathbf{S}_{t_1}) \log_2 \left(\frac{p(c_k \mid \mathbf{S}_{t_1})}{p(c_k \mid \mathbf{S}_{t_0})}\right) \tag{28}$$

4.2 Fitzgerald-Spielberger Draft Surplus Curves vs. Legacy Valuation Models

In the NFL draft trade simulator, the expected surplus value $S(p)$ of draft slot $p \in \{1, \ldots, 262\}$ is the discounted difference between expected on-field APY cap percentage and the 2011 CBA rookie wage scale cap hit $C_{\text{rookie}}(p, t)$ (with future picks discounted by $\delta = 0.90^k$ per year) (Ref. [A35, A62, A63, A64]):

$$S(p) = \sum_{t=1}^{4} \gamma^{t-1} \left(\mathbb{E}[\text{APY\%}_{p,t}] - C_{\text{rookie}}(p, t)\right) \tag{29}$$

Slot / Trade Case	Jimmy Johnson	Rich Hill (Market)	Harvard HSAC	Fitzgerald-Spielberger	Surplus Interpretation
Pick #1 (R1, P1)	3,000 (100%)	1,000 (100%)	494.6 (100%)	3,000 (100%)	High ceiling offset by highest rookie cap hit (Ref. [A62, A64]).
Pick #33 (R2, P1)	580 (19.3%)	180 (18.0%)	175.2 (35.4%)	1,228 (40.9%)	**Peak arbitrage**: $>2\times$ value vs. JJ/RH (Ref. [A63, A64]).
Pick #65 (R3, P1)	265 (8.8%)	78 (7.8%)	128.0 (25.9%)	885 (29.5%)	Day 2 retains $\sim 30\%$ of Pick #1 surplus (Ref. [A64]).
Pick #100 (R3/R4)	104 (3.5%)	36 (3.6%)	95.5 (19.3%)	671 (22.4%)	$6.4\times$ higher relative weight on FS than JJ (Ref. [A64]).

Slot / Trade Case	Jimmy Johnson	Rich Hill (Market)	Harvard HSAC	Fitzgerald-Spielberger	Surplus Interpretation
Pick #150 (R5)	31 (1.0%)	12 (1.2%)	63.7 (12.9%)	461 (15.4%)	Cheap depth contracts yield positive tail surplus (Ref. [A64]).
Pick #200 (R6)	11 (0.4%)	5 (0.5%)	40.6 (8.2%)	315 (10.5%)	Near-minimum wage preserves 10.5% value (Ref. [A64]).
Texans Trade-Up (#3)	—	+37% **Tax** (547 base)	—	+58% **Tax** (4,884 vs 3,089)	Texans paid 58% surplus tax for Will Anderson Jr. (Ref. [A64]).
Jets–Vikings Swap (#14+#143 for #23+#66+#86)	—	—	—	**Jets:** 119 / **MIN:** 184	Vikings gained +65 **surplus units** (≡ Pick #53) (Ref. [A62]).
Jamal Adams Trade	—	—	—	7.0% **Cap/Yr Surrendered**	3.34% + 1.50% + (3.53% · 0.9) − (1.11% · 0.9) (Ref. [A62]).

5 Phase 5: Serverless DuckLake Architecture, Zero-Local-Download Ingestion, and Traceable Hybrid RAG

5.1 The $0–$10/Month Serverless DuckLake Storage, Catalog, and Compute Topology

To host 659M+ play-by-play rows and optical tracking datasets at $0 to $10/month, the platform implements the **DuckLake** architecture (`ducklake.select`) across three tiers (Ref. [A61, A65, A66, A67, A68]):

1. **Columnar Storage Layer (Cloudflare R2):** Stores immutable ZSTD-compressed Apache Parquet files on R2 (10 GB free storage, 1M Class A writes/mo, 10M Class B reads/mo, $0/GB egress bandwidth), sorted hierarchically by `(season, game_id, team_id, player_id)` with small row groups (`ROW_GROUP_SIZE 32768` or `65536`) and `FLOAT32` downcasting so browser clients fetch kilobyte byte ranges via HTTP `Range` requests and predicate pushdown (Ref. [A57, A60, A65, A66, A69, A70, A71, A76, A77]).

2. **Transactional Metadata Catalog Layer (Neon Serverless Postgres / Cloudflare D1):** Replaces slow S3 Iceberg/Delta JSON manifest chains with a fast relational catalog on Neon Postgres or Cloudflare D1 (5 GB free SQLite storage, 100k writes/day free or 50M writes/mo on the $5 Workers Paid plan), completely bypassing DuckDB's single-writer file lock (Ref. [A65, A66, A67, A71, A78]).

3. **Client-Side OLAP & Edge Compute Layer (DuckDB-Wasm + Cloudflare Pages/Workers):** Spawns DuckDB-Wasm inside a browser Web Worker to query R2 Parquet files 10x faster than JS, feeding zero-copy Apache Arrow tables to SvelteKit + WebGL/Canvas visualizations (Ref. [A65, A72, A73, A74, A75]). Zero-local-download cloud-to-cloud ingestion runs inside GitHub Actions or Cloudflare Containers ($5/mo Workers Paid plan: 375 vCPU-min, 25 GiB-hrs RAM, 30s–5min CPU execution, 10M pooled monthly requests) using DuckDB `httpfs` to stream directly from SportsDataverse releases into R2, backed by a 3-tier anti-bot scraping stack (`python-mlb-statsapi` / ESPN JSON → `curl_cffi` TLS impersonation → `Camoufox` / CF Browser Rendering) (Ref. [A13, A19, A71, A79, A80, A81, A82, A83, A84, A85]).

4. **Real-Time WebSocket Quota Mitigation:** Production telemetry from October 10, 2026 demonstrates that SQLite-backed Durable Objects on the Workers Free tier can trigger `Error 1101: Exceeded allowed volume of requests in Durable Objects free tier` after 1,283,610 raw incoming WebSocket events + 44 HTTP invocations (64,224.5 billed requests under the 20:1 ratio, below the published 100k/day limit) (Ref. [A86]). The platform mitigates this on the $0 tier by coalescing live `diffPatch` updates into 5-second broadcasts over Supabase Realtime Broadcast (200 concurrent connections, 100 msgs/sec free) or by upgrading to the $5/mo Workers Paid plan (Ref. [A71, A86, A87, A88, A89, A90, A91]).

5.2 Traceable Three-Path Hybrid RAG Architecture

To guarantee zero-hallucination legal and statistical synthesis at $0.00/month, the platform combines Dense Semantic Retrieval via **Cloudflare Vectorize** (`@cf/baai/bge-base-en-v1.5`, 768-d, 5M dimensions free), Sparse Lexical Retrieval via **Cloudflare D1 SQLite FTS5 BM25** (`tokenize='porter unicode61'`), and Deterministic **DuckLake Text-to-SQL** tool execution, merging dense (r_{vec}) and sparse (r_{bm25}) ranks via Reciprocal Rank Fusion ($k = 60$) (Ref. [A13, A65, A66, A71, A90]):

$$\text{RRF_Score}(d) = \frac{1}{60 + r_{\text{vec}}(d)} + \frac{1}{60 + r_{\text{bm25}}(d)}, \quad \text{Provenance}(d) = \langle \text{doc_id}, \text{section_ref}, \text{r2_parquet_uri}, \text{snapshot_sha256}, \text{by} \tag{30}$$

6 Comprehensive Bibliography & Source Directory

Series A: Multi-Sport Quantitative Analytics Platform & DuckLake Sources

[A1] Baumer, B., et al. *Analyzing Baseball Data with R (3e): Statcast*. https://beanumber.github.io/abdwr3e/C_statcast.html

[A2] MLB Advanced Media. *Baseball Savant Statcast CSV Documentation (2026 ABS Update)*. https://baseballsavant.mlb.com/csv-docs

[A3] Fast, M. (2007). *Glossary of the Gameday Pitch Fields*. https://fastballs.wordpress.com/2007/08/02/glossary-of-the-gameday-pitch-fields/

[A4] Sabermetrics Community. *Statcast Vertical Break Measurement*. https://www.reddit.com/r/Sabermetrics/comments/ih69ue/statcast_vertical_break_measurement/

[A5] Sabermetrics Community. *Understanding pfx_z in Statcast*. https://www.reddit.com/r/Sabermetrics/comments/nentrm/can_someone_explain_what_pfx_z_is_from/

[A6] Nestico, T. J. *Creating the Perfect Pitching Summary*. https://medium.com/@thomasjamesnestico/creating-the-perfect-pitching-summary-7b8a981ef0c5

[A7] Kelly, J. *Stuff-Model: 12-Model Stratified Pitching Architecture*. https://github.com/jack-kelly-12/Stuff-Model

[A8] SABR. *2026 SABR Analytics Conference Research Abstracts (CGD & TrackMan xCTRL)*. https://sabr.org/analytics/2026-photos-2

[A9] GitHub Topics. *pybaseball Open-Source Ecosystem*. https://github.com/topics/pybaseball

[A10] Felton, T. D. *Optuna LightGBM Stuff+ Modeling & Validation*. https://tdfelton.github.io/

[A11] *xCTRL: Quantifying Pitcher Intent and Execution via Gaussian Mixture Models* (2025). https://arxiv.org/html/2508.19184v1

[A12] Prospects Live. *Application of Stuff Models on Triple-A Data*. https://www.prospectslive.com/an-introduction-to-the-application-of-stuff-models-on-triple-a-data/

[A13] Morishima, Y. *Park Factors, OAA, and Stuff+ in savant-extras v0.4.1*. https://dev.to/yasumorishima/park-factors-oaa-and-stuff-in-one-line-of-python-savant-extras-v032-v041-2hnd

[A14] Paraball Notes (2026). *Dollar Per WAR in 2025–2026 Free Agency*. https://www.paraballnotes.com/blog/dollar-per-war-2025-2026

[A15] SABR. *2027 SABR Analytics Conference Submission Guidelines*. https://sabr.org/analytics

[A16] SABR. *SABR 54 Call for Presentations*. https://sabr.org/latest/sabr-54-call-for-presentations/

[A17] SABR. *2025 SABR Analytics Conference Presentations*. https://sabr.org/analytics/presentations/2025

[A18] EdgeLab (2025). *The Best Free Sports Data APIs: A Developer's Review*. https://dev.to/edgelab/the-best-free-sports-data-apis-in-2025-a-developers-practical-review-14j9

[A19] PyPI. *python-mlb-statsapi v1.1.2 Documentation*. https://pypi.org/project/python-mlb-statsapi/

[A20] Gilani, S., et al. *The SportsDataverse GitHub Organization*. https://github.com/sportsdataverse

[A21] ClosingFive. *Box-Prior Impact Plus-Minus (IPM) Methodology*. https://closingfive.com/methodology

[A22] *Auditing Scorekeeper Bias and Stint Apportionment in NBA RAPM* (2026). https://arxiv.org/html/2605.24056v1

[A23] Basketball Analytics. *Open Source Data and RAPM Formulation in the NBA*. https://basketball-analytics.gitlab.io/rapm-data/open-source-data-nba.pdf

[A24] Squared2020 (2017). *Deep Dive on Regularized Adjusted Plus-Minus I*. https://squared2020.com/2017/09/18/deep-dive-on-regularized-adjusted-plus-minus-i-introductory-exam

[A25] Squared2020 (2018). *Regularized Adjusted Plus-Minus Part III*. https://squared2020.com/2018/12/24/regularized-adjusted-plus-minus-part-iii-what-had-really-happene

[A26] Annual Reviews of Statistics. *Statistical Modeling in Sports*. https://www.annualreviews.org/doi/pdf/10.1146/annurev-statistics-040720-015536

[A27] ResearchGate (2020). *An Extended Regularized Adjusted Plus-Minus (E-RAPM) Analysis for Lineup Management*. https://www.researchgate.net/publication/345490672

[A28] APBRmetrics Forum. *Prior-Informed RAPM and Ridge Shrinkage*. https://apbr.org/metrics/viewtopic.php?t=9844

[A29] OverTheApron. *NBA CBA Salary Matching Rules (2026–27)*. https://overtheapron.com/terms/salary-matching

[A30] ArmchairGM. *NBA First and Second Tax Aprons Guide*. https://armchairgm.io/guides/nba-first-second-tax-aprons

[A31] Seton Hall University Law Scholarship. *2023 NBA CBA Apron Analysis*. https://scholarship.shu.edu/cgi/viewcontent.cgi?article=2683&context=student_scholarship

[A32] Fanspo. *2026–27 NBA Trade Machine Rules & Thresholds*. https://fanspo.com/nba/trade-machine/rules

[A33] NBA CBA Analysis. *Specific Rules Around 2nd Apron Aggregation*. https://www.reddit.com/r/nba/comments/1g0mdwh/what_are_the_specific_rules_around_2nd_apron/

[A34] NBA CBA Analysis. *How the NBA Apron System Works*. https://www.reddit.com/r/nba/comments/1jjnjhl/eli5_how_does_the_apron_system_work/

[A35] Wharton Sports Analytics. *NBA & NFL Draft Value Curves*. https://wsb.wharton.upenn.edu/wp-content/uploads/2024/12/NBA_draft_curves-6.pdf

[A36] Svoboda, J. *NHL API Client Documentation*. https://github.com/jakubsvobodacz/nhl-api-client/blob/main/README.md

[A37] Zmalski. *NHL-API-Reference (api-web.nhle.com/v1)*. https://github.com/Zmalski/NHL-API-Reference

[A38] Tashlikovich, V. *NHL API: What Data Is Exposed and How to Analyse It.* https://medium.com/@vtashlikovich/nhl-api-what-data-is-exposed-and-how-to-analyse-it-with-python-7d5ac6838c2

[A39] Gilani, S., et al. *fastRhockey v1.0.0 CRAN Release Notes.* https://fastrhockey.sportsdataverse.org/news/index.html

[A40] R-Universe. *fastRhockey Package Repository.* https://sportsdataverse.r-universe.dev/fastRhockey

[A41] SportsDataverse. *sportsdataverse-py Python Package.* https://github.com/sportsdataverse/sportsdataverse-py

[A42] CRAN. *fastRhockey v1.0.0 Reference Manual.* https://cran.r-project.org/web/packages/fastRhockey/readme/README.html

[A43] GitHub. *sportsdataverse/fastRhockey Source Repository.* https://github.com/sportsdataverse/fastRhockey

[A44] Zmalski. *NHL Edge Endpoints in api-web.nhle.com/v1 (Issue #69).* https://github.com/Zmalski/NHL-API-Reference/issues/69

[A45] fastRhockey. *nhl_edge_team_shot_speed_detail.R Implementation.* https://github.com/sportsdataverse/fastRhockey/blob/main/R/nhl_edge_team_shot_speed_detail.R

[A46] rdrr.io. *nhl_edge_team_detail Documentation.* https://rdrr.io/cran/fastRhockey/man/nhl_edge_team_detail.html

[A47] Hockey Analytics Community. *NHL API Roster Endpoint Caveats.* https://www.reddit.com/r/hockey/comments/1njsyxs/nhl_api/

[A48] Reign Creative. *Expected Goals (xG) and GSAx in Hockey.* https://reigncreativellc.com/blog/what-is-expected-goals-in-hockey/

[A49] NHL Hockey Betting Tips. *Goalie GSAx Evaluation.* https://nhlhockeybettingtips.com/goalies/

[A50] MetricGate. *Expected Goals (xG) with XGBoost.* https://metricgate.com/docs/expected-goals-xg-xgboost/

[A51] Sofianakos, N. *NHL Expected Goals (xG) Model Engineering.* https://medium.com/@nsofianakos/nhl-expected-goals-xg-model-39bd2edba932

[A52] PMC. *Spatio-Temporal Features in Ice Hockey Shot Modeling.* https://pmc.ncbi.nlm.nih.gov/articles/PMC11524524/

[A53] Evolving-Hockey. *Reviving Regularized Adjusted Plus-Minus for Hockey.* https://evolving-hockey.com/blog/reviving-regularized-adjusted-plus-minus-for-hockey/

[A54] *Spatial Shot Quality in Ice Hockey* (2025). https://arxiv.org/html/2512.00203v2

[A55] *Hierarchical Bayesian Shooter Finishing Effects in Hockey xG* (2025). https://arxiv.org/html/2511.23072v1

[A56] Aalto University. *Machine Learning Models for Ice Hockey xG.* https://aaltodoc.aalto.fi/bitstreams/ea245658-fc00-4f84-97ca-1959b5da154a/download

[A57] PyPI. *nfl-data-py v0.3.3 Documentation.* https://pypi.org/project/nfl-data-py/

[A58] SMU Data Science Review. *NFL Big Data Bowl Coordinate Standardization.* https://scholar.smu.edu/cgi/viewcontent.cgi?article=1249&context=datasciencereview

[A59] UC eScholarship. *Spatio-Temporal Tracking Analysis in Professional Football.* https://escholarship.org/content/qt5qw5382d/qt5qw5382d_noSplash_983c1629d166d6b046b4a2d65e75e816.pdf

[A60] DuckDB Documentation. *Cloudflare R2 Import & S3 Compatibility.* https://duckdb.org/docs/2.0/guides/network_cloud_storage/cloudflare_r2_import

[A61] SportsDataverse. *Open-Source Sports Data Portal.* https://www.sportsdataverse.org/

[A62] Open Source Football (2023). *NFL Draft Value Chart & Surplus Analysis.* https://opensourcefootball.com/posts/2023-02-23-nfl-draft-value-chart/

[A63] PickSwap. *NFL Draft Value Charts (JJ, Rich Hill, HSAC, FS).* https://pickswapapp.com/draft-value-charts.html

[A64] A to Z Sports. *Assessing NFL Draft Pick Exchanges with Fitzgerald-Spielberger and Rich Hill Charts.* https://atozsports.com/nfl-draft/two-best-most-accurate-nfl-draft-trade-value-charts-assessing-pi

[A65] Blankenship, G. *Cloudflare Free & Paid Developer Tier Architecture.* https://gist.github.com/garyblankenship/27a4c57eca4aa5d659ee3c509668b66d

[A66] DuckLake. *DuckLake Specification and FAQ.* https://ducklake.select/faq

[A67] Tobilg. *The Age of $10-a-Month Lakehouses (DuckLake + R2).* https://tobilg.com/posts/the-age-of-10-dollar-a-month-lakehouses/

[A68] Hacker News. *DuckLake Architecture Discussion.* https://news.ycombinator.com/item?id=47761997

[A69] DuckDB v1.5 Docs. *Cloudflare R2 Storage Integration.* https://duckdb.org/docs/1.5/guides/network_cloud_storage/cloudflare_r2_import

[A70] BlazingCDN. *Cloudflare Pricing for Developers: Workers, Pages & R2.* https://blog.blazingcdn.com/en-us/cloudflares-pricing-for-developers-a-closer-look-at-workers-page

[A71] NateCue. *Cloudflare Workers Paid $5 Plan Limits & Containers.* https://www.natecue.com/en/learn/tech/cloudflare-workers-paid-5-dollar-plan/

[A72] Casper, M. *duckdb-query-r2 Reference Implementation.* https://github.com/casperm/duckdb-query-r2

[A73] Hacker News. *DuckDB-Wasm In-Browser Analytics Benchmarks.* https://news.ycombinator.com/item?id=34741195

[A74] DuckDB Docs. *httpfs and S3 API Extension.* https://duckdb.org/docs/2.0/core_extensions/httpfs/s3api

[A75] Hacker News. *Zero-Copy Arrow and DuckDB Streaming.* https://news.ycombinator.com/item?id=48553388

[A76] DuckDB LTS Docs. *Parquet Row Group Size & Predicate Pushdown Tips.* https://duckdb.org/docs/lts/data/parquet/tips

[A77] StackOverflow. *Predicate Pushdown in DuckDB over HTTP Range Requests.* https://stackoverflow.com/questions/76696239/predicate-pushdown-in-duckdb-for-a-parquet-file-in-s3

[A78] Hacker News. *DuckDB Concurrency & Single-Writer Patterns.* https://news.ycombinator.com/item?id=40864091

[A79] Clay.fyi. *Cheap Lakehouses with Parquet, DuckDB, and R2.* https://clay.fyi/blog/cheap-opentelemetry-lakehouses-parquet-duckdb-iceberg/

[A80] Gilani, S. *fastRhockey Data Engineering Notes.* https://www.saiemgilani.com/notes/fastrhockey

[A81] VHub Systems (2026). *Web Scraping Tools Comparison 2026: curl_cffi vs Camoufox vs Playwright.* https://dev.to/vhub_systems_ed5641f65d59/web-scraping-tools-comparison-2026-requests-vs-curlcffi-vs-playw

[A82] PublicAPIs. *ESPN Sports API Documentation.* https://publicapis.io/espn-sports-api

[A83] Pseudo-R. *Public-ESPN-API Reference.* https://github.com/pseudo-r/Public-ESPN-API

[A84] ToddRob99. *MLB-StatsAPI Endpoints Wiki (diff-Patch).* https://github.com/toddrob99/MLB-StatsAPI/wiki/Endpoints

[A85] Hockey Community. *CapWages and PuckPedia Cap Data Sources.* https://www.reddit.com/r/hockey/comments/1e1va2j/capwages_looking_like_its_going_to_be_a_good_alt/

[A86] Cloudflare Community (Oct 2026). *Durable Objects Free Quota Block Below 100k After WebSocket 20:1 Adjustment.* https://community.cloudflare.com/t/durable-objects-free-quota-block-below-100k-after-websocket-20-1967650

[A87] AI Agency Plus. *Supabase Free Tier Realtime Limits.* https://aiagencyplus.com/supabase-free-tier-limits/

[A88] Ably. *Supabase Realtime vs Managed WebSockets.* https://ably.com/compare/supabase-vs-twilio

[A89] RapidDevelopers. *Supabase Broadcast Debouncing Architecture.* https://www.rapidevelopers.com/lovable-prompts/lovable-prompts-for-building-chat-application

[A90] Cloudflare Developers. *Workers Platform Limits (CPU, Memory, Bundles).* https://developers.cloudflare.com/workers/platform/limits/

[A91] Shattered.io (2026). *Cloudflare Workers vs Fastly Compute Pricing 2026.* https://shattered.io/cloudflare-workers-vs-fastly-compute-pricing-2026/

Series B: Baseball Latent State-Space, Park Factor, and Projection Literature

[B1] `PROMPT_continue_contact_quality.md` (Internal State-Space Contact Quality Specification).

[B2] `PROMPT_research_park_babip.md` (Internal Component Park Factor & BABIP Audit).

[B3] `BLOCKED_ON_USER.md` / Chadwick Bureau Baseball Register: https://github.com/chadwickbureau/register

[B4] `Improving Player Projections.md` (Pitcher Kinematics, Censoring & Convex Valuation).

[B5] `PROMPT_claude_modeling_round3.md` (VectorSSM Multivariate State-Space Formulation).

[B6] `PROMPT_claude_continue.md` (MiLB Translation & WAR Replacement Accounting).

[B7] `PROMPT_glm_agent_round3.md` & Tango, T. (2004). *Marcel the Monkey Forecasting System.* http://www.tangotiger.net/marcel/

[B8] Jensen, S. T., Shirley, K. E., & Wyner, A. J. (2009) / Brown, L. D. (2008). *Hierarchical Bayesian Modeling of Hitting.* https://arxiv.org/html/0902.1360v1

[B9] MLB Glossary. *Steamer Projection System.* https://www.mlb.com/glossary/projection-systems/steamer

[B10] FanGraphs Prep. *Build and Test Your Own Projection System.* https://blogs.fangraphs.com/fangraphs-prep-build-and-test-your-own-projection-system/

[B11] FanGraphs Library. *Projection Systems Principles.* https://library.fangraphs.com/principles/projections/

[B12] Chapman University Digital Commons. *Novel Statistical and ML Methods for Forecasting.* https://digitalcommons.chapman.edu/cgi/viewcontent.cgi?article=1010&context=cads_dissertations

[B13] FanGraphs. *Where ZiPS and Steamer Disagree.* https://blogs.fangraphs.com/where-zips-and-steamer-disagree/

[B14] Pitcher List. *Fantasy 101: Projections and Benchmarks.* https://pitcherlist.com/fantasy-101-projections-and-benchmarks/

[B15] FanGraphs Prep. *Regression Towards the Mean.* https://blogs.fangraphs.com/fangraphs-prep-regression-towards-the-mean/

[B16] Szymborski, D. (2014). *10 Lessons I Have Learned About Creating a Projection System.* https://tht.fangraphs.com/10-lessons-i-have-learned-about-creating-a-projection-system/

[B17] Szymborski, D. *2023 ZiPS Projections: Atlanta Braves.* https://blogs.fangraphs.com/2023-zips-projections-atlanta-braves/

[B18] Szymborski, D. *ZiPS zStats for Hitters at the Midpoint.* https://blogs.fangraphs.com/zips-zstats-for-hitters-at-the-midpoint/

[B19] Szymborski, D. *The 2022 ZiPS Projections Are Coming!.* https://blogs.fangraphs.com/the-2022-zips-projections-are-coming/

[B20] Szymborski, D. *Hitter zStats Entering the Homestretch, Part 1 (Validation).* https://blogs.fangraphs.com/hitter-zstats-entering-the-homestretch-part-1-validation/

[B21] Szymborski, D. *zStats for Hitters, June Update.* https://blogs.fangraphs.com/zstats-for-hitters-june-update/

[B22] Wikipedia. *PECOTA.* https://en.wikipedia.org/wiki/PECOTA

[B23] SportsPundit. *Player Empirical Comparison and Optimization Test Algorithm.* https://sportspundit.com/baseball/terms/2180-player-empirical-comparison-and-optimization-test-algorithm/

[B24] CalState ScholarWorks. *Simulation-Based Projections for Baseball Statistics.* https://scholarworks.calstate.edu/downloads/8p58pg38j

[B25] Silver, N. *Baseball Prospectus Basics: The Science of Forecasting.* https://www.baseballprospectus.com/news/article/2659/baseball-prospectus-basics-the-science-of-forecasting/

[B26] Baseball Prospectus. *10-Year Projections, UPSIDE, Percentiles and Comparables.* https://www.baseballprospectus.com/news/article/23016/baseball-prospectus-news-10-year-projections-upside-percentiles-

[B27] Baseball Prospectus. *Glossary: PECOTA.* https://legacy.baseballprospectus.com/glossary/?context=6&category=true

[B28] Baseball Prospectus (2025). *PECOTA 2025: An Introduction.* https://www.baseballprospectus.com/news/article/96300/pecota-week-pecota-2025-an-introduction/

[B29] Baseball Prospectus. *Reintroducing PECOTA: The Seven Percent Solution.* https://www.baseballprospectus.com/news/article/12133/reintroducing-pecota-the-seven-percent-soluti

[B30] FantasyPros (2026). *Most Accurate Fantasy Baseball Projections (2025 Results).* https://www.fantasypros.com/2026/02/most-accurate-fantasy-baseball-projections-2025-results/

[B31] Carty, D. *THE BAT X Projections! THE BAT plus Statcast.* https://rotogrinders.com/pages/the-bat-x-projections-the-bat-plus-statcast-3361569

[B32] Carty, D. *Introducing THE BAT X.* https://fantasy.fangraphs.com/introducing-the-bat-x/

[B33] SoxMachine (2026). *A Look at THE BAT X 2026 Projections.* https://soxmachine.com/2026/01/a-look-at-the-bat-x-2026-chicago-white-sox-projections

[B34] RotoGrinders. *THE BAT X Projection System by Derek Carty.* https://rotogrinders.com/the-bat

[B35] Carty, D. *Introducing THE BAT X for Pitchers and THE BATcast Stuff Model.* https://blogs.fangraphs.com/introducing-the-bat-x-for-pitchers-and-the-batcast-stuff-model/

[B36] FanGraphs. *New Hitter xBABIP Based on BIS Batted Ball Data.* https://fantasy.fangraphs.com/new-hitter-xbabip-based-on-bis-batted-ball-data/

[B37] Carleton, R. A. / Birdland Metrics. *A Guide to Statistical Stabilization and Regression.* https://birdlandmetrics.com/articles/stabilization-regression

[B38] FanGraphs. *Giancarlo Stanton Is Putting on a Fireworks Show.* https://blogs.fangraphs.com/giancarlo-stanton-is-putting-on-a-fireworks-show/

[B39] FanGraphs. *You Can't Fake Exit Velocity.* https://blogs.fangraphs.com/you-cant-fake-exit-velocity/

[B40] `PROMPT_glm_agent.md` (MLB Stats API MiLB Ingestion & Savant FRV Protocol).

[B41] Birdland Metrics. *The Pete Alonso Effect.* https://birdlandmetrics.com/articles/pete-alonso-impact

[B42] FanGraphs. *Musings at the Intersection of Launch Angle Consistency and Hard-Hit Rate.* https://blogs.fangraphs.com/musings-at-the-intersection-of-launch-angle-consistency-and-

[B43] Chamberlain, A. *Let's Talk About Launch Angle Tightness.* https://fantasy.fangraphs.com/lets-talk-about-launch-angle-tightness/

[B44] Pitcher List. *Hitting with a TON of Authority.* https://pitcherlist.com/hitting-with-a-ton-of-authority/

[B45] SABR Journal. *The Many Flavors of DIPS: A History and an Overview.* https://sabr.org/journal/article/the-many-flavors-of-dips-a-history-and-an-overview/

[B46] Appelman, D. *Park Factors Added to FanGraphs Guts!.* https://blogs.fangraphs.com/park-factors-added-to-fangraphs-guts/

[B47] Sabermetrics Community. *Graded MLB Projection Model.* https://www.reddit.com/r/Sabermetrics/comments/1asrij2/built_a_public_graded_mlb_projection_model/

[B48] Taylor & Francis (2018). *Machine Learning Applications in Baseball: A Systematic Literature Review.* https://www.tandfonline.com/doi/pdf/10.1080/08839514.2018.1442991

[B49] Grokipedia. *Adjusted Batting Average.* https://grokipedia.com/page/adjusted_batting_average

[B50] Baseball Community. *fWAR Comparisons and Park Adjustments.* https://www.reddit.com/r/baseball/comments/1n074j9/nick_kurtz_and_juan_soto_both_have_the_same_fwar/

[B51] FanGraphs. *2013 Park Factors Updated.* https://blogs.fangraphs.com/2013-park-factors-updated/

[B52] Albert, J., et al. *Handbook of Statistical Methods and Analyses in Sports.* https://api.pageplace.de/preview/DT0400.9781498737371_A30123636/preview-9781498737371_A30123636.pdf

[B53] Mitchell, C. (2015). *KATOH: Forecasting Major League Hitting with Minor League Stats.* https://tht.fangraphs.com/katoh-forecasting-a-hitters-major-league-performance-with-minor-league-stats/

[B54] Wikipedia. *Clay Davenport.* https://en.wikipedia.org/wiki/Clay_Davenport

[B55] Baseball Wiki. *Baseball Prospectus.* https://baseball.fandom.com/wiki/Baseball_Prospectus

[B56] Davenport, C. *DTs vs. MLEs - A Validation Study.* https://www.baseballprospectus.com/news/article/49/dts-vs-mles-a-validation-study/

[B57] Mitchell, C. *A Primer on a New and Improved KATOH.* https://blogs.fangraphs.com/a-primer-on-a-new-and-improved-katoh/

[B58] Mitchell, C. *FanGraphs Community Research Archive.* https://community.fangraphs.com/author/mitchell/page/2/

[B59] Kalman, R. E. (1960) / MDPI (2025). *State-Space Systems & Uncertainty-Aware Forecasting.* https://www.mdpi.com/2078-2489/16/12/1016

[B60] `ROUND4_STEPS.md` (Adaptive Gaussian Comps Kernel & Role-Split RAR Specification).

[B61] Sarris, E., & Bay, M. *Stuff+, Location+, and Pitching+ Primer.* https://library.fangraphs.com/pitching/stuff-location-and-pitching-primer/

[B62] FanGraphs. *PitchingBot and Stuff+ Pitch Modeling on FanGraphs.* https://blogs.fangraphs.com/pitchingbot-and-stuff-pitch-modeling-are-now-on-fangraphs/

[B63] FanGraphs. *Minors to the Majors: Calculating Individual Pitch Grades.* https://fantasy.fangraphs.com/minors-to-the-majors-calculating-individual-pitch-grades/

[B64] Szymborski, D. *Pitcher zStats Entering the Homestretch, Part 1 (Validation).* https://blogs.fangraphs.com/pitcher-zstats-entering-the-homestretch-part-1-validation/

[B65] Appelman, D. (2016). *Park Factors - 5 Year Regressed.* https://library.fangraphs.com/park-factors-5-year-regressed/

[B66] Morris, C. N. (1983). *Parametric Empirical Bayes Inference: Theory and Applications.* https://www.researchgate.net/profile/Carl-Morris-3

[B67] Jensen, S. T., et al. (2009). *Hierarchical Bayesian Modeling of Hitting Performance in Baseball.* https://faculty.wharton.upenn.edu/wp-content/uploads/2009/11/Shanejensen.traj09.pdf

[B68] Klinger, T. *State-Space Parameter Estimation and Covariance Filtering.* https://dgk.badw.de/fileadmin/user_upload/Files/DGK/docs/c-787.pdf